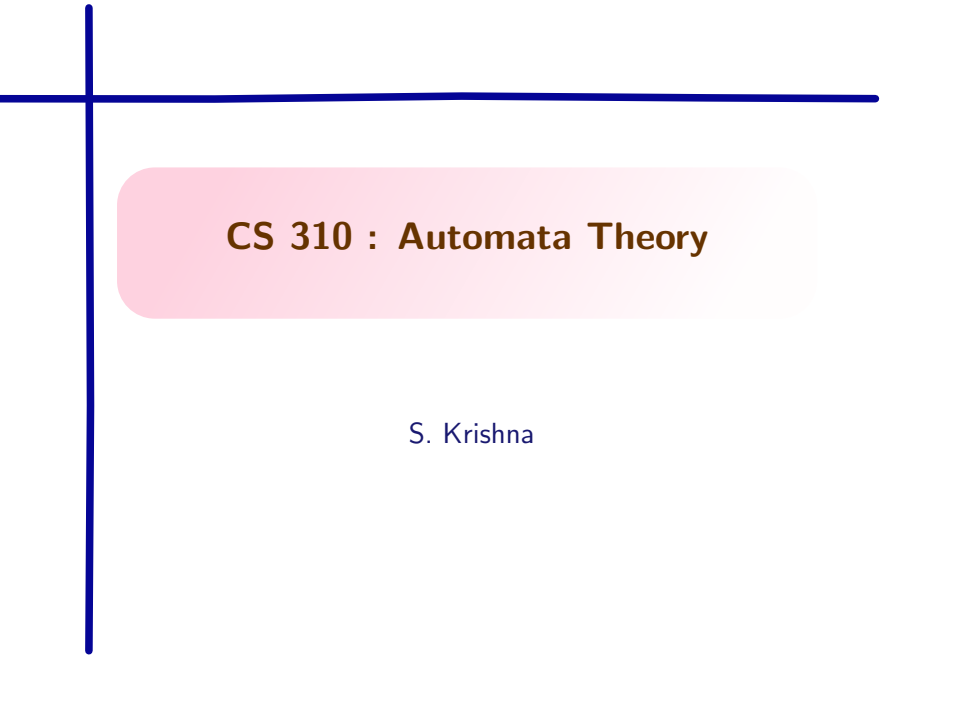


A decorative blue crosshair consisting of a vertical line and a horizontal line intersecting at the top-left corner of the slide.

CS 310 : Automata Theory

S. Krishna

A necessary and sufficient Pumping Lemma

(Jaffe, 1978)

Jaffe's Pumping Lemma

Theorem

$L \subseteq \Sigma^*$ is regular **if and only if** there exists $n \geq 1$ such that: for every word $y \in \Sigma^n$, there exist words u, v, w factorizing y as $y = uvw$ with $v \neq \varepsilon$, and for **every** $z \in \Sigma^*$ and every $i \geq 0$,

$$uv^i wz \in L \iff uvwz \in L.$$

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- ▶ L regular $\Rightarrow$ pumping property holds
- ▶ pumping property holds $\Rightarrow L$ regular: build a DFA directly from the property.

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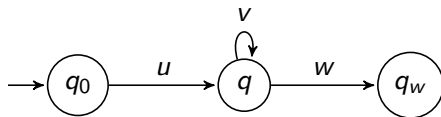
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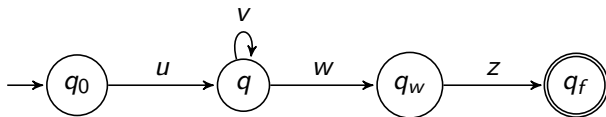
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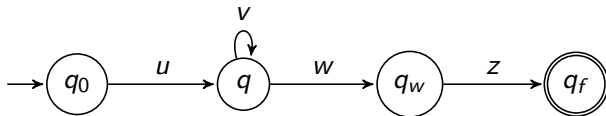
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The loop on q means u and uv, uv^i all reach the *same* state. $uwz, uvwz, uv^2wz$ are all either accepted or not for any continuation z .

Sufficiency: property holds $\Rightarrow L$ regular

Let L be a language.

Assume $\exists n \geq 1$ such that for all words $y \in \Sigma^n$ there exist words u, v, w factorizing $y = uvw$ with $v \neq \epsilon$ and for all words $z \in \Sigma^*$,

$$uvwz \in L \Leftrightarrow uv^i wz \in L$$

for all $i \geq 0$. Then L is regular.

Construct a DFA M for L .

An Example

$L = \{w \in \{a, b\}^* : w \text{ contains } ab \text{ as a subword}\}$. We know L is regular.

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 - ▶ $v = b$ ($u = a, w = \varepsilon$): $uwz = a \notin L$.
 - ▶ $v = ab$ ($u = \varepsilon, w = \varepsilon$): $uwz = \varepsilon \notin L$.
 - ▶ So $n \neq 2$.

An Example

$L = \{w \in \{a, b\}^* : w \text{ contains } ab \text{ as a subword}\}$

- ▶ Consider $n = 3$. $\Sigma^3 = \{aaa, aab, aba, abb, baa, bab, bba, bbb\}$.
- ▶ For each $y \in \Sigma^3$, we can check if the property holds.

Checking all 8 words of length 3

y	u	v	w	uw	uvw	$uv^i w \ (i > 1)$
aaa						
aab						
aba						
abb						
baa						
bab						
bba						
bbb						

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y	u	v	w	uw	uvw	$uv^i w \ (i > 1)$
aaa	a	a	a	aa	aaa	a^{i+2}
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aba	ab	a	ϵ	ab	aba	$ab \cdot a^i$
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Building the automaton

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- ▶ Note that $|uw| \leq 2$
- ▶ We can start building the automaton by first generating words of length < 3 , and use the above property to generate words of length 3
- ▶ Then use the same property to generate words of length > 3

Building the automaton

- ▶ Words of length < 3 : $\epsilon, a, b, aa, bb, ba, ab$.
- ▶ Consider words of length 3

Building the automaton

- ▶ Words of length < 3 : ϵ , a , b , aa , bb , ba , ab .
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So $x = aaaa$ behaves exactly like aa : $xb \in L \iff aab \in L$.

Sliding the window along $x = aabab$

a*abab*

so far: *a* (keep growing)

Sliding the window along $x = aabab$

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so far: *a* (keep growing)

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aabab → **ab**ab

window *aab* → *ab*

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so far: *aa* (keep growing)

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abab → **abb**

window *aba* → *ab*

abb → **ab**

window *abb* → *ab*

Sliding the window along $x = aabab$

aabab so far: *a* (keep growing)

aabab so far: *aa* (keep growing)

aabab $\rightarrow$ **ab**ab window *aab* $\rightarrow$ *ab*

abab $\rightarrow$ **abb** window *aba* $\rightarrow$ *ab*

abb $\rightarrow$ **ab** window *abb* $\rightarrow$ *ab*

We reduce to *ab*

Sliding the window along $x = bbaaa$

bbaaa

so far: b (keep growing)

Sliding the window along $x = bbaaa$

bbaaa

so far: *b* (keep growing)

bbaaa

so far: *bb* (keep growing)

Sliding the window along $x = bbaaa$

bbaaa so far: b (keep growing)

bbaaa so far: bb (keep growing)

bbaaa $\rightarrow$ **ba**aa window $bba \rightarrow ba$

Sliding the window along $x = bbaaa$

$bbaaa$ so far: b (keep growing)

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$bbaaa \rightarrow bbaa$ window $bba \rightarrow ba$

$baaa \rightarrow aaaa$ window $baa \rightarrow aa$

Sliding the window along $x = bbaaa$

bbaaa so far: *b* (keep growing)

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bbaaa → **ba**aa window *bba* → *ba*

baaa → **aa**a window *baa* → *aa*

aaa → **aa** window *aaa* → *aa*

Sliding the window along $x = bbaaa$

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$bbaaa \rightarrow bbaa$ window $bba \rightarrow ba$

$baaa \rightarrow aaaa$ window $baa \rightarrow aa$

$aaa \rightarrow aa$ window $aaa \rightarrow aa$

We reduce to aa

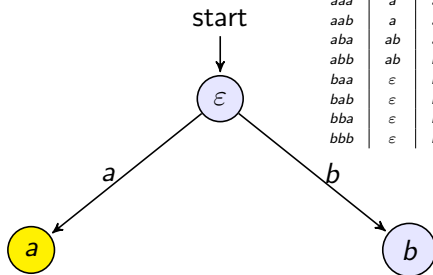
Building the automaton

start



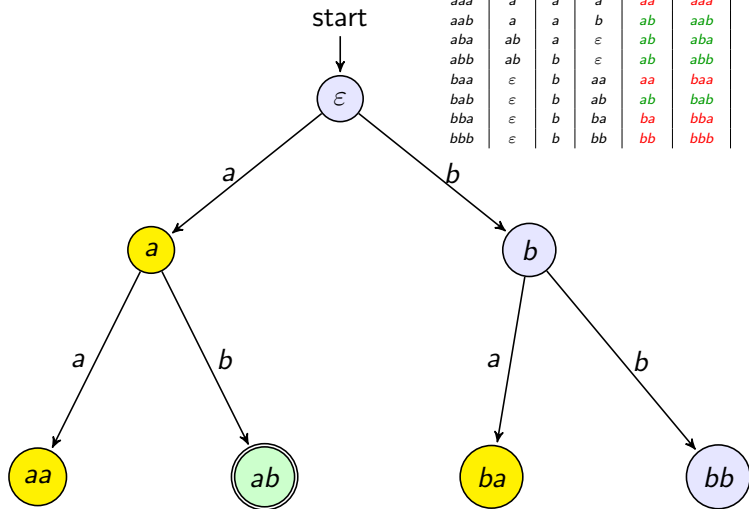
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baa	ϵ	b	aa	aa	baa	$b^i aa$
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Building the automaton



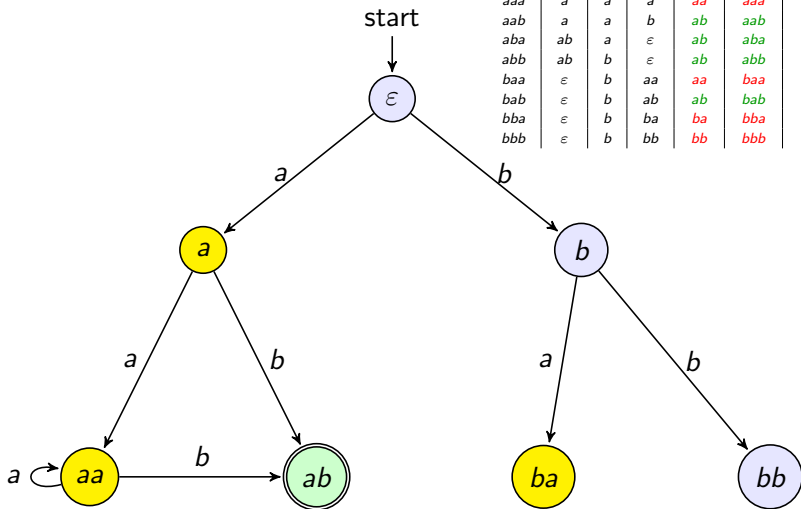
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Building the automaton



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aaa	a	a	a	aa	aaa	a^{i+2}
aab	a	a	b	ab	aab	$a^i \cdot ab$
aba	ab	a	ϵ	ab	aba	$ab \cdot a^i$
abb	ab	b	ϵ	ab	abb	ab^{i+1}
baa	ϵ	b	aa	aa	baa	$b^i aa$
bab	ϵ	b	ab	ab	bab	$b^i ab$
bba	ϵ	b	ba	ba	bba	$b^{i+1} a$
bbb	ϵ	b	bb	bb	bbb	b^{i+2}

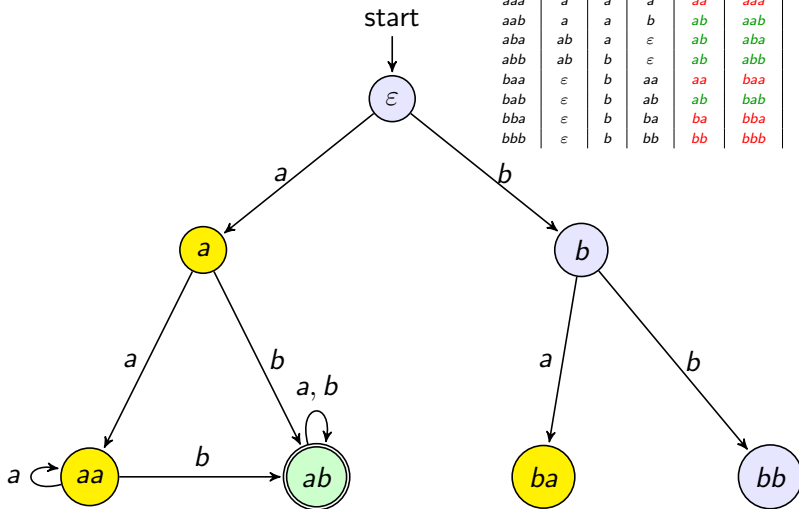
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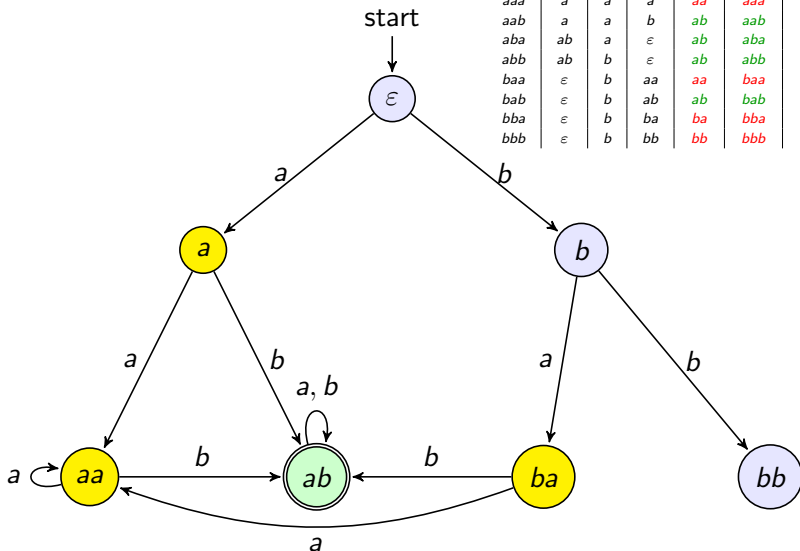
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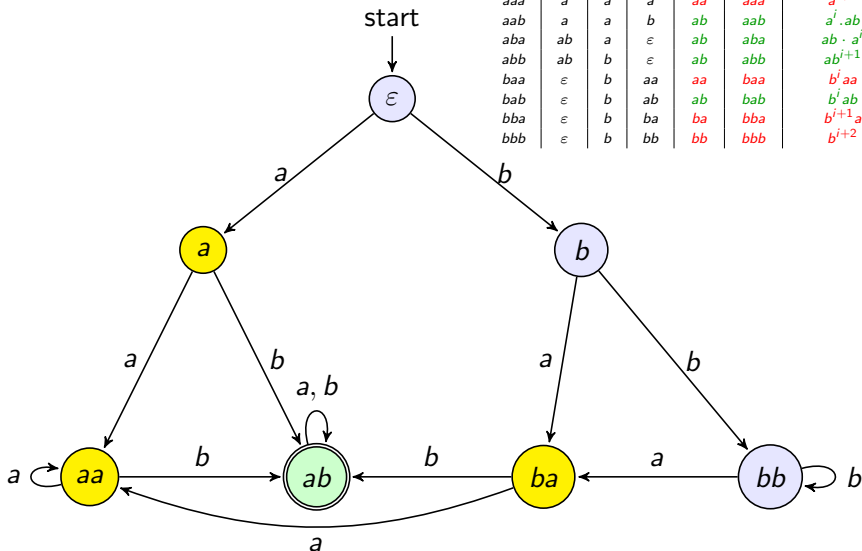
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To do : define transitions δ , and prove $L(M) = L$.

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- ▶ If $|sa| < n$, define $\delta(s, a) = sa$. Clearly, $sa \in Q$ by definition.
- ▶ Now consider $|sa| = n$. Let $y = sa \in \Sigma^n$.
 - ▶ We have u, v, w factorizing y satisfying $uv^i wz \in L \Leftrightarrow uvwz \in L$ for all z .
 - ▶ For each such y , precompute u, v, w .
 - ▶ Define $\delta(s, a) = uw$. $uw \in Q$, $|uw| < n$ (recall that v is a pump)

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Set $z = \varepsilon$ and take $x = w$: then $w \in L \iff s \in L$

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► Case $|sa| < n$: new state $s \cdot a$. For any z :

$$(x \cdot a) \cdot z \in L \iff x \cdot (a \cdot z) \in L \stackrel{\text{IH}}{\iff} s \cdot (a \cdot z) \in L \iff (s \cdot a) \cdot z \in L.$$

Invariant preserved.

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- ▶ Case $|sa| = n$: let $y = s \cdot a = u \cdot v \cdot w$, new state $u \cdot w$. For any z :

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- ▶ **Necessity** : L regular $\Rightarrow$ every long witness has a decomposition against every continuation z .

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- ▶ **Necessity** : L regular $\Rightarrow$ every long witness has a decomposition against every continuation z .
- ▶ **Sufficiency** : given the decomposition rule, the DFA M built accepts L .
So L is regular.

Jaffe's property holds $\iff L$ is regular